DAT321/DIT847: Software Quality

Welcome to the examination for the *Software Quality* course. Each question has a number of points assigned shown in the square brackets. When the question is broken down into smaller sub-questions the part of the points for that specific sub-questions are also shown as following:

1. *[10 pts].*
    a. *[2 pts]*
    b. *[8 pts]*

The percentage of points and the corresponding grade is presented below (100 points in total):

% of points	DAT321	DIT847
[0, 50%)	U	U
[50%, 65%)	3	G
[65%, 85%)	4	G
[85%, 100%]	5	VG

Instructions about how to submit this exam are found in the Assignments section of Canvas (under the Written Exam Submission page). You must submit a PDF file with your answers. Make sure that you label your answers with the question numbers so we can identify which question your answer is referring to. You can choose any editor to write your answers.

Important: You must write clear, readable, understandable and unambiguous answers. An advice is to refer to the provided software product context used in this exam. Whenever possible refer to modules, features and information about the software product or software development process described in the context.

The questions in this exam refer to the **ISO 25010:2011** that categorises internal and external software quality attributes into eight characteristics, as well as the quality in use characteristics.

Questions about the exam contact:
Francisco Gomes, tel. 031 772 69 51, francisco.gomes@cse.gu.se

The exam review will be done via Zoom scheduled for:
Date: 2021-01-22 between 09:00 – 10:30.
Zoom Link: https://chalmers.zoom.us/j/62891716138
Password: 317535

The questions in this exam are related to the context below. Note that you **must** justify your exam answers with i) the theory and terminology from software quality and ii) their connections to the elements in this context (e.g., teams, product, processes, architecture, customers, etc.).

You are hired by a media streaming company Stream-Co that offers the most popular **media streaming services on the Internet.** The software product for this exam is a real-time streaming platform where end-users can watch several movies and series. Examples of similar software products are Netflix, HBO Nordic, HBO Max, Prime Video, Disney+, etc.

Features, business and customers:
In order to access the software, users must first login to access the media catalogue made available by Stream-Co. When first creating an account, users must choose a subscription plan with different payment models (e.g., monthly, yearly) where the user provides credit card or banking information to Stream-Co. Upon successful login, the UI Platform (**Module A**) shows the media available to the user so that they can begin to watch their favourite media. The UI Platform is installed in the user's client side and is built to allow access from various hardware devices, such as a laptop's browser, mobile phones tablets or even directly via Smart TVs.

A Subscription platform (M**odule D**) handles all storage and access to user's account information such as e-mail, name, home address, phone number. The billing of accounts is handled by another software provided by a partner company named Bank-Co. In order to ensure communication, Stream-Co uses a Billing platform (**Module E**) able to communicate with the software developed by Bank-Co and bill its user.

Stream-Co also uses a Streaming platform (**Module B**) from which the media data is sent over the internet to the user's screens. Since users have different quality of internet connections, the Streaming platform automatically adapts the quality of the video to better match the user's connection bandwidth. In the background, a Recommender Platform (**Module C**) monitors the streaming data and the users' choices in order to learn and understand viewing patterns of the user and recommend new series and movies by using different machine learning algorithms.

Design and architecture:
The software product uses a typical Client-server architecture (Figure 1), and users watch live content using an Internet connection. Each module's responsibilities are listed below; they are connected to the features of the system listed above. These modules exchange information and also depend on each other for the streaming service to function properly.

Modules A is deployed in the client side of the architecture, whereas Modules B, C and D are deployed in the server side of the architecture. Module F is its own platform developed and deployed by Bank-Co. Our only responsibility is to ensure that the interfaces between the Billing and Subscription platforms can communicate to know which payments were successful.

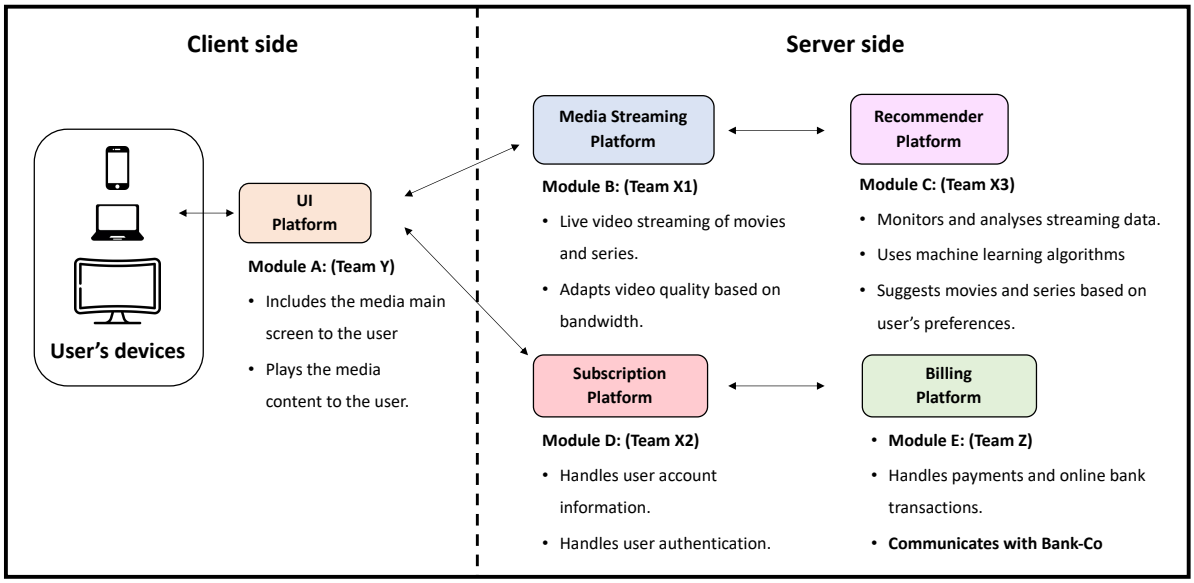

Figure 1 – Summary of Modules and Features of the Stream-Co Product

Development process and teams:

There are three teams in Stream-Co. Team X is a team of 30 engineers divided into smaller teams of roughly equal size that work with the server side of the service (Modules B, C and D). Teams X1 and X2 work on the Streaming and Subscription platforms, respectively. They are composed of a mix between junior and senior architects, developers and testers experienced with agile methods. Team X3 is new to Stream-Co and is composed of junior developers, two senior software engineers and one data scientist experienced with machine learning. Most members of Team X3 have not worked together and did not use agile methods before.

Team Y is composed of 10 engineers working in the client side of the service (Module A) and is composed of developers, testers and one user experience (UX) designer. Finally, Team Z is a small team of 2 engineers responsible for maintaining the communication with Bank-Co (Module E). All members of Teams Y and Z have worked with agile development before.

Each team has their own isolated development environment using version control. However, only Team X3 is **not** using continuous integration (CI), such that the necessary tool chain for CI is not installed in their development environments.

Software Product – Future plans:

Stream-Co is aiming to expand and include its own social network platform in the application. Via this social network, users that watch the same videos can share messages and add each other as contacts. In order to save time and costs with software development, the Social Network Platform (Module F) aims to reuse the Subscription and Recommender modules. Therefore, Stream-Co will soon hire engineers for Team S to develop Module F.

1. **[25 pts]** Using the context above, answer the following questions:

   a. [15 pts] Choose **three** software product quality's characteristics, **and, for each,** provide examples of why they are relevant to the software product at Stream-Co. When justifying your choices, you **must** use the features and modules listed in Figure 1 as sources for your examples.

   b. [10 pts] Choose **two** quality in use characteristics and explain which modules or features we could use to measure those chosen characteristics.

2. **[15 pts]** Stream-Co wants to foster sustainability in ICT. Using the Sustainability Awareness Framework, you must (i) choose or suggest **three product features** and explain the (ii) immediate, (iii) enabling and (iv) structural effects of those features. You must also indicate the **corresponding dimension** in which those effects occur.

   For this question you are free to choose to modify an existing feature (from Figure 1) or create a new feature. We refer to feature as a functionality in our software product. Make sure to clearly distinguish between the immediate, enabling and structural effects in your answer.

3. **[15 pts]** Considering any of the features or modules in Figure 1, choose three different levels of testing and, for each level chosen, provide at least two examples of test cases.

4. **[15 pts]** Considering the decision to introduce Continuous Integration to Team X3, answer the following questions:

   a. [5 pts] What is the necessary development toolchain that Team X3 needs to have in order to properly use Continuous Integration.

   b. [10 pts] List **two activities** you would do or teach to Team X3 before introducing CI. Justify why those activities would be relevant to introduce CI.

5. **[20 pts]** Considering internal/external quality measures and quality attributes, answer the following questions:

   a. [5 pts] An analysis of technical debt revealed that all modules need refactoring and have the same debt, such that restoring the maintainability of each module would take 10 days. You only have resources to choose one module to reduce this technical debt. **Which module** would you choose and **why**?

b. [5 pts] You were asked to design a two-dimensional chart (x-axis and y-axis) to visualise the internal quality of the different modules in Figure 1. Which two quality measures would you show in this chart? Which quality attributes would be easier to understand using your proposed visualisation?

c. [10 pts] Provide **two examples** of external quality measures for the software product. For each example, you must explain which module or feature from Figure 1 they would be associated with.

6. **[10 pts]** The lead tester in Team X1 wants to improve the quality of all server-side modules by comparing two new code reviewing techniques named Technique A and Technique B. The goal is identifying which technique is better at finding more code smells. She then asks all junior and senior developers to use both techniques and measures the number of code smells found in the code. This process is repeated for Modules B and C. An example of the data collected during the experiment is shown below:

Engineer ID	Experience	Module	Technique	Number of Code Smells
P1	Senior	Module C	Technique A	8
P1	Senior	Module C	Technique B	7
P1	Senior	Module B	Technique A	15
P1	Senior	Module B	Technique B	20
P2	Junior	Module C	Technique A	4
P2	Junior	Module C	Technique B	8
P2	Junior	Module B	Technique A	10
P2	Junior	Module B	Technique B	12
…				

Write down the **mathematical model definition** for this experiment using **any** variable names and priors of your choice. State the ontological and epistemological reasons for your likelihood. Remember to clearly state and justify the choices and assumptions regarding your model.